



**OUTCOMES-BASED EDUCATION COURSE DESIGN
COURSE SYLLABUS**

1. COURSE SPECIFICATIONS

Course Title	Automata Theory and Formal Languages	Course Code:	CS302
Course Description	This course introduces the fundamental concepts of automata theory and formal languages, focusing on the mathematical models of computation and their applications. Topics include finite automata (FA), deterministic finite automata (DFA), nondeterministic finite automata (NFA), the equivalence of DFA and NFA, regular grammars, and regular languages. Students will learn to design, analyze, and implement automata-based solutions for pattern recognition and language processing problems. Emphasis is placed on understanding the relationships between these models, their expressive power, and their role in computer science applications such as lexical analysis, text processing, and simple language recognition. At the end of the course, the students must design and implement a lexical analyzer or parser using concepts from automata theory and formal languages.		
SY/Semester	2025-2026/1 st Semester	Credit Units	3 units = 3 hours Lec/week
Program(s) to which it contributes	BSCS – 3	Prerequisite/ Corequisite	CS204

2. VISION AND MISSION OF NwSSU

VISION	MISSION
The premier technological university in the region providing transformative education where graduates are globally competitive, innovative, and responsive to the demands of a changing world. (BOR No. 101 s. 2022)	NwSSU shall lead in providing highly technical and professional education and lifelong learning in the trade, fishery, agriculture, science, education, commerce, engineering, forestry, nautical, and other emerging programs in the digital age. It shall generate cutting-edge technology and undertake sustainable community development in accordance with the university mandates, thrusts, and directions. (BOR No. 101 s. 2022)





3. INSTITUTIONAL GRADUATE OUTCOMES (IGO)

NwSSU IGO	Operational Indicators	
IGO1. Creative and Critical Thinkers	☐ Solves problems ☐ Generates new and innovative ideas ☐ Exhibits questioning mind	☐ Receptive to new ideas ☐ Makes a sound judgment
IGO2. Life-long Learners	☐ Manifests willingness to learn ☐ Updates knowledge, skills and qualification ☐ Accepts challenges while calculating risks	☐ Applies theory to practice ☐ Influences transformative change
IGO3. Effective Communicators	☐ Demonstrates good and oral and written communication ☐ Expresses thoughts clearly and responsibly	☐ Listens attentively and emphatically ☐ Responds to social issues and problems
IGO4. Morally and Socially Upright Individuals	☐ Leads a morally and socially upright life ☐ Demonstrates ethics and professionalism ☐ Responds to community development needs	☐ Advocates environmental conservation and cultural preservation ☐ Manifests gender sensitivity ☐ Shows social maturity in dealing with others

4. PROGRAM GRADUATE OUTCOMES (PGO)

NWSSU-IGO	PGO Code	BSCS Program Graduate Outcomes
IGO1. Creative and Critical Thinkers	CS01	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
	CS02	Identify, analyze, formulate, research literature, and solve complex computing problems and requirements reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
	CS03	An ability to apply mathematical foundations, algorithmic principles and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices.
	CS04	Knowledge and understanding of information security issues in relation to the design, development and use of information systems.
	CS05	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
	CS06	Create, select, adapt and apply appropriate techniques, resources and modern computing tools to complex computing activities, with an understanding of the limitations to accomplish a common goal.





NWSSU-IGO	PGO Code	BSCS Program Graduate Outcomes
IGO3. Effective Communicator	CS07	Function effectively as an individual and as a member or leader in diverse teams in multidisciplinary setting by being able to express opinions rationally, assertively but not arrogantly, respecting other's opinions and to listen attentively and empathetically
	CS08	Expresses ideas and feelings accurately and in a clearly organized manner – in both writing and speaking with the computing community and with society at large about complex computing activities by being able to discern and processes information objectively, write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
IGO4. Morally and Socially Upright Individuals	CS09	The ability to recognize the legal, social, ethical and professional issues involved in the utilization of computer technology, be guided by the adoption of appropriate professional, ethical and legal practices, to serve others with compassion, lead and exemplary life, and to live out the Via, Veritas, Vita and become an image of God's justice.
IGO2. Life-long Learner	CS10	Recognize the need and have the ability to engage in independent learning for continuous development as a computing professional, to value all forms of learning and to always strive for excellence.

5. COURSE LEARNING OUTCOMES

LEARNING OUTCOMES At the end of the course, the students should be able to:	CS01	CS02	CS03	CS04	CS05	CS06	CS07	CS08	CS09	CS10
LO1. Explain the fundamental concepts of automata theory and formal languages, including their classifications, significance, and applications in computer science.	2	2	2	2	2	2	3	3		
LO2. Construct finite automata (FA), deterministic finite automata (DFA), and nondeterministic finite automata (NFA) to represent and recognize specific languages or patterns.	2	2	2	2	2	2	3	3		
LO3. Demonstrate the equivalence of DFA and NFA through formal conversion methods.	2	2	2	2	2	2	3	3		
LO4. Develop regular grammars to define regular languages, and analyze their properties and expressive limitations.	3	3	3	3	3	3	3	3		
LO5. Apply automata and grammar concepts to design and implement solutions for pattern recognition and simple language processing problems.	3	3	3	3	3	3	3	3		

1-Introduced

2-Reinforced

3- Practiced





6. MAJOR COURSE OUTPUTS (MCO)/OTHER REQUIREMENTS AND GRADING SYSTEM

LEARNING OUTCOMES	REQUIRED OUTPUT	DUE DATE
LO1, LO2, LO3, LO4, LO5	Design and implementation of a lexical analyzer or parser using concepts from automata theory and formal languages MCO1. Project Proposal. Students are required to submit a project proposal that addresses the problem stated above, demonstrating effective teamwork.	Week 8
LO1, LO2, LO3, LO4, LO5	MCO2. Oral Defense/Presentation. Students shall present a fully functional version of the proposed project that meets all specific requirements outlined in the project proposal, accompanied by complete documentation.	Week 16

OTHER REQUIREMENTS:

Aside from the major course outputs stated above, the following learning activities are required:

Assessment Activity	Learning Outcomes it Represents
Examinations	LO1, LO2, LO3, LO4, LO5
Class Standing (Quizzes, Group Work, Participation, Attendance, Recitation)	LO1, LO2, LO3, LO4, LO5

CLASSROOM POLICIES:

- ✓ Access the course Moodle page **CS302: AUTOMATA THEORY AND FORMAL LANGUAGES** _____ (SY 2025-2026) at moodle.nwssu.edu.ph.
- ✓ Join our group chat in messenger.
- ✓ The discussion area of the course (FB group/Group chat) is reserved for postings related to course work only. Postings of a personal or nonacademic nature are not permitted and may be removed by the instructor.
- ✓ There is to be no food or drink allowed in the computer lab.
- ✓ Cell phone use during class is prohibited. You will receive one warning before any disciplinary action is taken.
- ✓ The computers and internet are to be used responsibly and for the sole purpose of work related to the class. Inappropriate use of the computer will result in disciplinary action
- ✓ Your workstation will be kept and left for the next class in a clean and orderly manner.
- ✓ Cheating is not allowed.
- ✓ Programming exercises are individual work.





- ✓ **Deadline for the submission of MCO1 is before the Midterm Examination and MCO2 is before the Final Examination**

GRADING SYSTEM:

This course shall use Grading Category B.

To pass this course, one must accumulate at least 75 points through the course requirements discussed above. The maximum points that a student can obtain through each requirement are shown below:

Requirement/Assessment Task	Weight
Major course output	40%
Major exams	30%
Class standing	30%
TOTAL	100%

7. LEARNING PLAN

LO	Course Contents / Subject Matter	Teaching and Learning Activities (Methodology)	Assessment	Resources	Timeframe
	Preliminaries and Orientation: Orientation of Vision, Mission, and Objectives of the University/College Course Orientation Classroom Policies	Direct Instruction. Discussion on the content of the syllabus.	Class Participation	Moodle LMS, laptop/cell phone, syllabus	Week 1
LO1	Module 1: Introduction to Automata Theory & Formal Languages - Introduction to Automata Theory - Applications of Automata Theory - Basic Fundamentals/Central Concepts of Automata Theory - Formal Languages and Chomsky Hierarchy	✓ Direct Instructions. Discussion on the basic concepts of Automata Theory and the Chomsky Hierarchy. ✓ Multimedia presentation ✓ Q&A ✓ Think-pair-share activity on real-world applications.	- Paper and Pencil Test - Class Participation - Oral recitation	Moodle LMS, laptop/cell phone, pictures and video clips from various media, and references stipulated in the syllabus (see item 8).	Week 2-3





LO	Course Contents / Subject Matter	Teaching and Learning Activities (Methodology)	Assessment	Resources	Timeframe
LO2	Module 2: Finite Automata - Introduction to Finite Automata - Structural Representations - Acceptance of Strings and Languages by FA	✓ Direct Instructions. Discussion about Finite Automata, its components, and structural representation using transition diagram and transition tables. ✓ Step-by-step demonstration ✓ Guided problem solving ✓ Individual and group exercises on FA design.	- Paper and Pencil Test - Seatwork - Graded recitation.	Moodle LMS, laptop/cell phone, pictures and video clips from various media, and references stipulated in the syllabus (see item 8).	Week 4-6
SUBMISSION/PRESENTATION OF MCO 1					Week 7-8
MIDTERM EXAM					Week 9
LO2, LO3	Module 3: DFA and NFA - Deterministic Finite Automata (DFA) - Nondeterministic Finite Automata (NFA) - Equivalence of DFA and NFA	✓ Lecture with worked examples ✓ Small group workshops to perform conversions ✓ Peer review of solutions.	- Paper and Pencil Test - Problem set on DFA–NFA and its conversion	Moodle LMS, laptop/cell phone, pictures and video clips from various media, and references stipulated in the syllabus (see item 8).	10-12
LO4	Module 4: Regular Expressions and Languages - Introduction to Regular Expressions - Finite Automata and Regular Expressions	Direct Instructions. Discussion about regular expressions Individual Work. The students will solve some computing problems to be presented in the class. Group Work/Brainstorming. The students will be divided into groups to create a solution to	- Paper and Pencil Test - Class Participation - Oral recitation - Hands-on exercises	Moodle LMS, laptop/cell phone, pictures and video clips from various media, and references stipulated in the syllabus (see item 8).	Week 13-15





LO	Course Contents / Subject Matter	Teaching and Learning Activities (Methodology)	Assessment	Resources	Timeframe
		a problem using regular expressions			
LO5	SUBMISSION/PRESENTATION OF MCO 2				Week 16
	FINAL EXAM				Week 17
	CLASS INTEGRATION (returning the examination booklets/quizzes/activities, feed backing, etc.)				Week 18

8. REFERENCES/SUGGESTED READINGS

Books:

1. Carpon, Cristy (2023). CS302 – Automata Theory and Formal Languages Multimedia Material.
2. Linz, P. (2016). An Introduction to Formal Languages and Automata (6th ed.). Jones & Bartlett Learning.
3. Sipser, M. (2012). *Introduction to the Theory of Computation* (3rd ed.). Cengage Learning.

Internet Resources:

4. Automata Simulator – <https://automatonsimulator.com>
5. Hughes, Charles (2014). Formal Languages and Automata Theory. <https://www.pdfdrive.com/formal-languages-and-automata-theory-e15426187.html>
6. Hussain, A. (n.d). Theory of Automata/Computation, Formal Languages, DFA, NFA, CFG, CFL, Pushdown Automata, Turing Machine, Undecidability. Retrieved from <https://www.udemy.com/course/theory-of-automata-and-computation/learn/lecture/24676592?start=0#overview>
7. Linz, Peter. An Introduction to Formal Languages and Automata Theory 5th Edition. Retrieved from <https://fall14cs.files.wordpress.com/2017/04/an-introduction-to-formal-languages-and-automata-5th-edition-2011.pdf>
8. Madana Mohana, R. (n.d.). Introduction to Automata Theory, Languages and Computation. Retrieved from <https://www.udemy.com/course/formal-languages-and-automata-theory-e/learn/lecture/25409914#overview>
9. MIT OpenCourseWare: Theory of Computation – <https://ocw.mit.edu>
10. Raj, B. (2019, August 12). *Finite automata and regular expressions*. GeeksforGeeks. <https://www.geeksforgeeks.org/finite-automata-and-regular-expressions>
11. Sipser, Michael (2013). Introduction to Theory of Computation 3rd edition. Retrieved from https://notendur.hi.is/mae46/Haskolinn/5.%20misseri%20-%20Haust%202018/Formleg%20ma%CC%81%20og%20reiknanleiki/Introduction%20to%20the%20theory%20of%20computation_third%20edition%20-%20Michael%20Sipser.pdf
12. Shahi, Tej (2013). Introduction to Automata Theory, Formal Language and Computability Theory. <https://www.pdfdrive.com/introduction-to-automata-theory-formal-language-and-computability-theory-e37220113.html>
13. Sunitha, KVN & Kalyani, N. (2016). Formal languages and Automata Theory. Retrieved from <http://libgen.rs/search.php?req=Formal+Languages+and+Automata+Theory&open=0&res=25&view=simple&phrase=1&column=def>
14. TutorialsPoint. (n.d.). *Automata theory*. https://www.tutorialspoint.com/automata_theory/index.htm





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Base Syllabus Information	
Syllabus Author	Cristy S. Carpon, MSIT
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